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USING AI-PREDICTING BIKE USAGE AND OPTIMIZING OPERATIONS AT REPAIR IN BIKE SHARING SYSTEMS

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229Y1A3934

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Department : Computer Science and Engineering

B. Tech : Computer Science and Engineering

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Guide : Mrs.Swetha

Abstract

- Supply chain responsiveness and Big Data Analytics (BDA) have become essential in managing modern transportation systems that generate large and complex datasets. Bike-sharing systems (BSS) continuously produce data related to user behavior, weather conditions, time, and bike availability. Managing this data effectively is crucial for predicting bike demand and ensuring timely repair and maintenance operations. Traditional forecasting and supply chain methods struggle to handle such data complexity and dynamic demand patterns.
- This study focuses on improving supply chain responsiveness in bike-sharing systems by integrating machine learning techniques with supply chain network design. Using publicly available bike-sharing data from Washington D.C., various machine learning models such as neural networks, decision tree regression, K-nearest neighbors, support vector machines, and random forest are applied to predict bike usage and repair needs. A capacity extension-based supply chain model is developed and optimized using a gradient-based search method. The results demonstrate improved operational flexibility, reduced repair lead time, and increased system throughput.

Existing System

- Existing bike-sharing systems lack accurate demand forecasting and flexible repair capacity planning, leading to increased repair lead times and reduced system throughput. There is a need for an integrated approach that combines machine learning-based prediction with responsive supply chain network design to handle big data challenges effectively.

Exiting Gap

- Traditional forecasting models fail to capture complex demand patterns
- Limited integration between machine learning predictions and supply chain design
- Fixed repair shop capacity reduces responsiveness
- Insufficient use of big data analytics for operational decision-making

Scope and Limitations

Scope

- Use of real-world bike-sharing data
- Application of multiple machine learning models
- Optimization of repair operations through capacity extension

Limitations

- Dataset limited to a single city (Washington D.C.)
- Real-time data integration is not considered
- Computational complexity increases with large datasets

OBJECTIVES

- To analyze bike-sharing system data using big data analytics
- To predict bike usage and repair needs using machine learning models
- To compare the performance of different prediction algorithms
- To improve repair shop responsiveness through capacity extension
- To reduce repair lead time and increase system throughput

Proposed System

- The project collects publicly available bike-sharing data and preprocesses it to handle missing values and inconsistencies.
- Machine learning algorithms such as neural networks, decision tree regression, KNN, SVM, and random forest are trained to predict bike demand.
- Based on predicted demand, a supply chain network model for bike repair operations is developed.
- Capacity extension is allowed to improve flexibility, resulting in a nonlinear optimization problem solved using a gradient-based search method.

Tools and Technologies

Software Requirements:

Operating system : Windows 7 Ultimate

Coding Language : Python

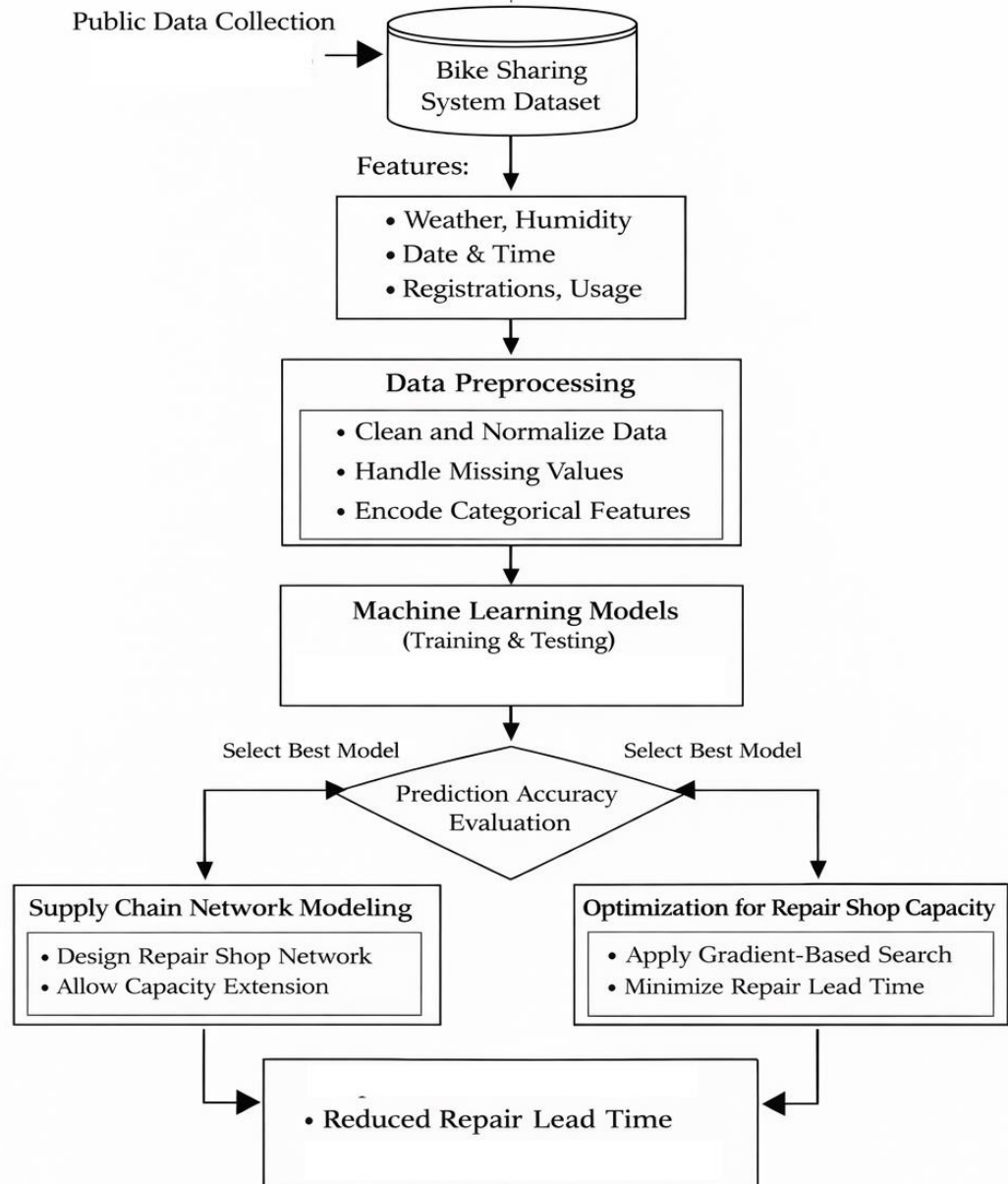
Additional Python Packages:

Scikit-learn

Django

Pandas

Architecture



Algorithm

Input:

Bike Sharing System Dataset D
(weather, humidity, date, time, registrations, usage data)

Output:

Predicted bike demand
Optimized repair shop capacity

Step 1: Data Collection

Load public BSS dataset D (Washington D.C.)

Step 2: Data Preprocessing

Remove duplicate records, Handle missing values,
Normalize numerical attributes, Encode categorical variables

Step 3: Feature Extraction

Extract features:

- Weather conditions, - Temperature and humidity
- Date and time attributes, - User registration details
- Historical bike usage

Step 4: Dataset Splitting

Split dataset D into:

Training set (D_train)

Testing set (D_test)

Step 5: Model Training

Train machine learning models using D_train

Step 6: Prediction

For each trained model:

Predict bike usage on D_test

Step 7: Model Evaluation

Calculate performance metrics:

- Prediction accuracy

Select best-performing model M_best

Step 8: Supply Chain Network Modeling

Design bike repair supply chain network

Model repair shop capacity as a variable

Step 9: Decision Making

Allocate repair capacity based on predicted demand

Step 11: Output Results

Report: - Reduced repair lead time End Algorithm

Modules

Admin Module

The Admin module is responsible for managing the entire system, controlling data flow, and monitoring model performance.

1. Admin Login

- Secure authentication (username & password)

- Access control to prevent unauthorized usage

2. Dataset Management

- Upload social media datasets (CSV, JSON, etc.)

- View, edit, and delete datasets

- Preprocess data (remove noise, stopwords, emojis)

4. Model Training

- Train ML/NLP models such as:

 - ML models

6. Reports & Analytics

- Generate reports on:

 - Model performance

 - User activity

Modules

User Registration & Login

Sign up with basic details

Secure login system

2. Text Input

Enter Bike Repairs text manually

3. Text Analysis

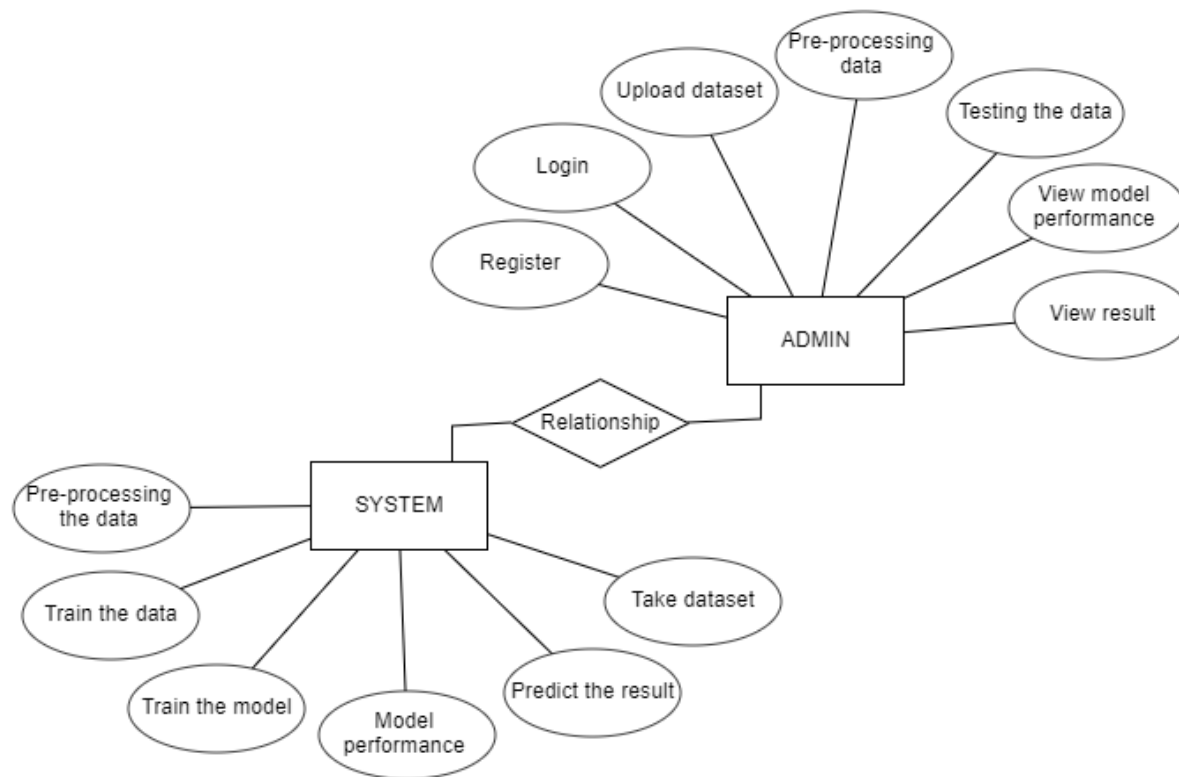
System processes input using ML techniques

4. Prediction Result

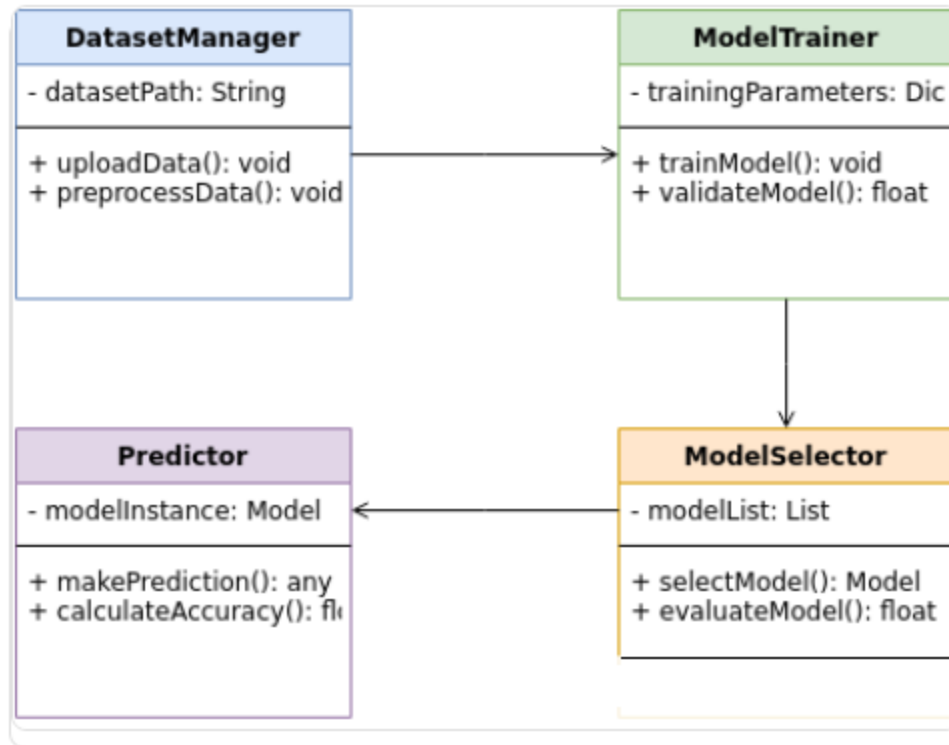
Displays Predicted Results

System Design

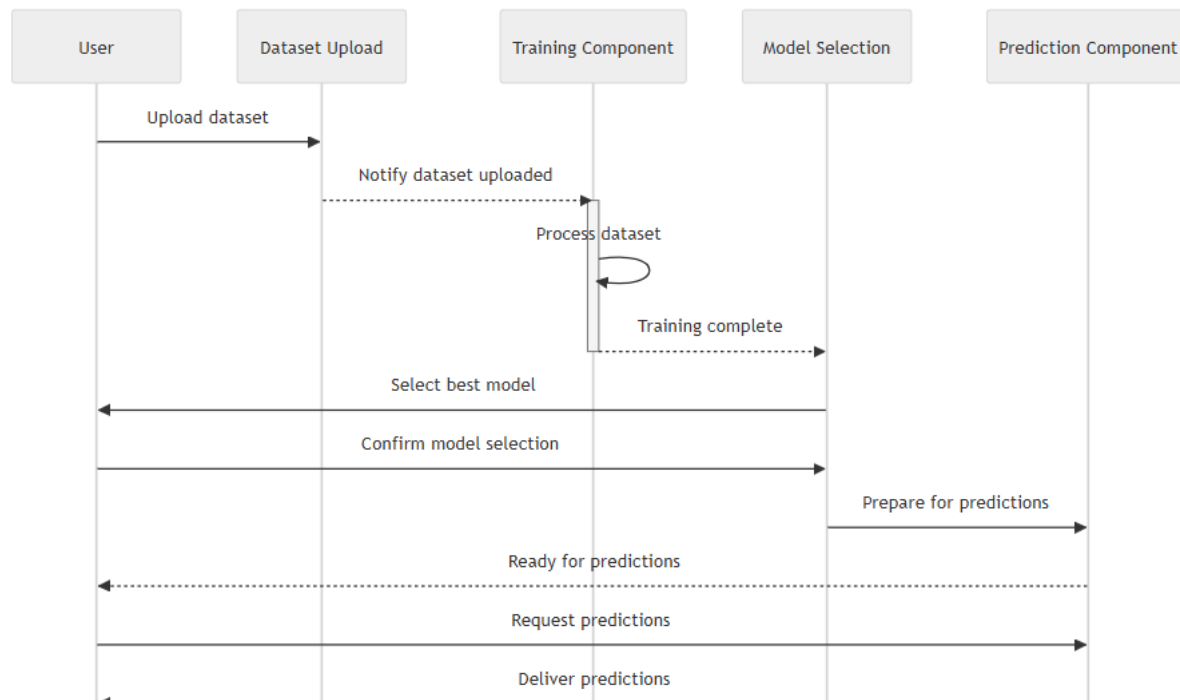
- ER Diagram



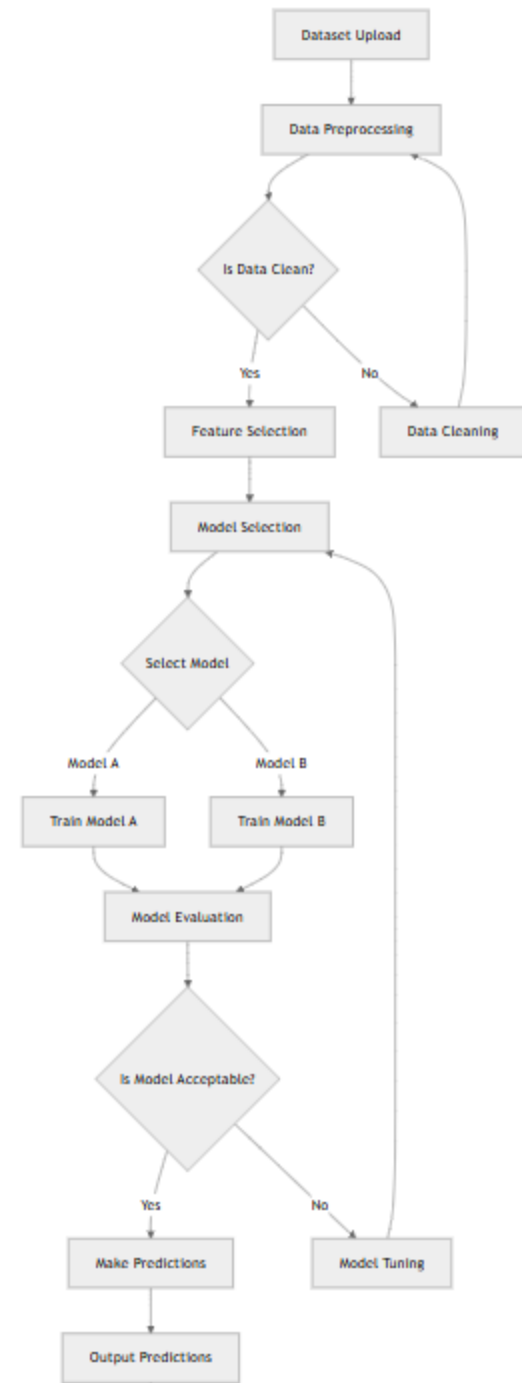
Class Diagram



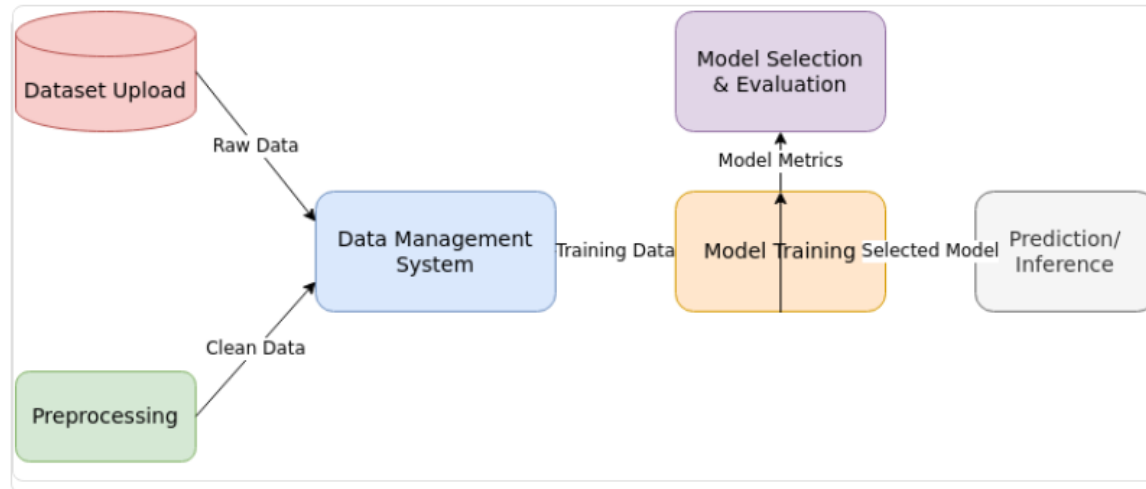
Sequence Diagram



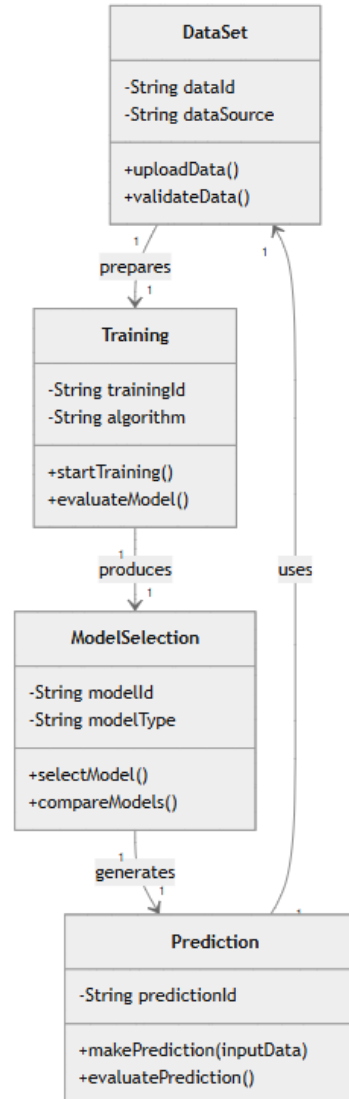
Activity Diagram



COLLOBORATION IDAGRAM



Deployment diagram

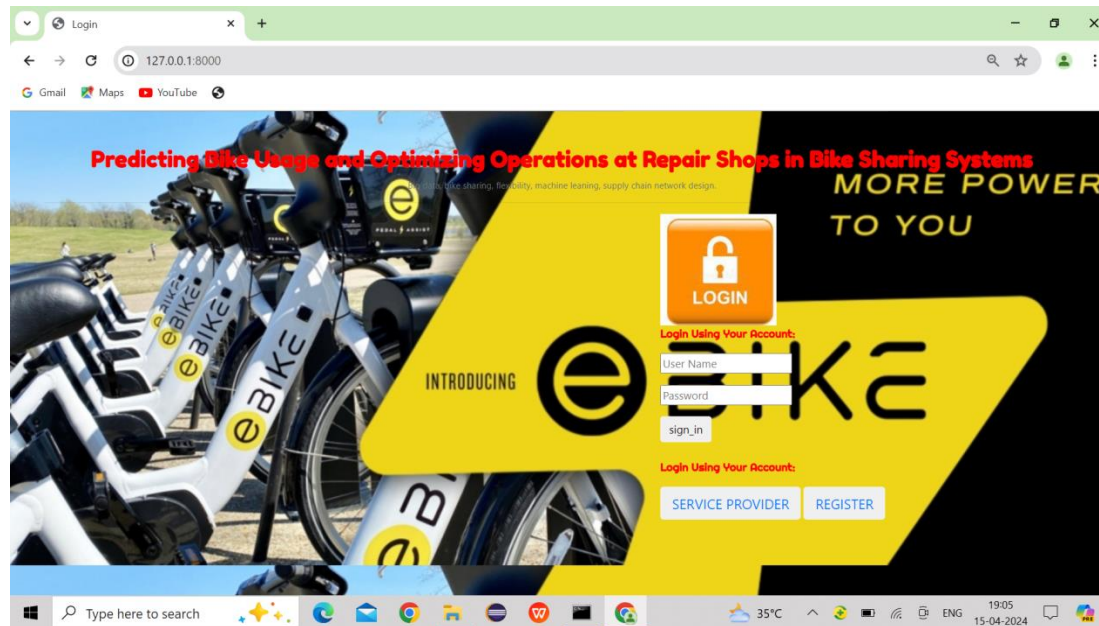


Results

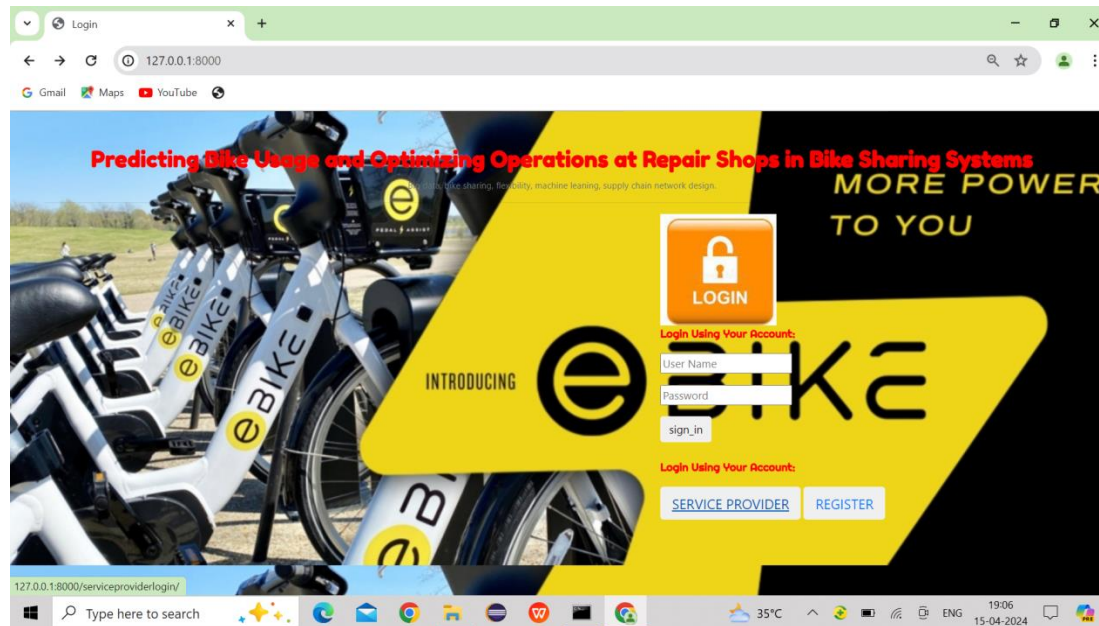
Data Sets

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Fid	Age_band	Sex_of_dri	Education	Vehicle_dr	Driving_ex	Type_of_ve	Owner_of	Service_ye	Defect_of	Road_allig	Road_surfs	Road_surfs	Light_con
2	172.217.1	Over 51	Male	Junior hig	Owner	5-10yr	Automobil	Owner	Above 10yr	No defect	Tangent ro Asphalt ro	Wet or dan	Daylight	
3	10.42.0.21	18-30	Male	Junior hig	Employee	2-5yr	Lorry (41?	Government	2-5yrs	No defect	Tangent ro Asphalt ro	Wet or dan	Darkness	
4	203.205.1	18-30	Male	Elementar	Employee	Below 1yr	Other	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Wet or dan	Darkness	
5	10.42.0.21	Over 51	Male	Junior hig	Employee	2-5yr	Pick up up	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
6	10.42.0.21	Over 51	Male	Junior hig	Employee	5-10yr	Lorry (11?	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
7	192.229.1	31-50	Male	High scho	Employee	Above 10yr	Motorcycle	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
8	10.42.0.21	31-50	Male	High scho	Employee	Above 10yr	Pick up up	Owner	1-2yr	No defect	Tangent ro Asphalt ro	Dry	Daylight	
9	180.149.1	31-50	Male	Junior hig	Employee	5-10yr	Lorry (11?	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
10	10.42.0.21	18-30	Male	Elementar	Employee	1-2yr	Lorry (11?	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
11	10.42.0.21	18-30	Male	Elementar	Employee	5-10yr	Automobil	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Darkness	
12	211.236.2	Over 51	Male	Junior hig	Employee	5-10yr	Automobil	Owner	Above 10yr	No defect	Tangent ro Asphalt ro	Dry	Darkness	
13	10.42.0.21	Unknown	Female	Junior hig	Employee	1-2yr	Other	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Darkness	
14	10.42.0.21	31-50	Male	Junior hig	Owner	5-10yr	Other	Owner	Unknown	No defect	Tangent ro Other	Dry	Darkness	
15	172.217.1	31-50	Male	High scho	Owner	Above 10yr	Automobil	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Darkness	
16	10.42.0.21	Unknown	Female	Writing &	Owner	5-10yr	Other	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
17	10.42.0.21	18-30	Male	Above high	Employee	Above 10yr	Taxi	Government	5-10yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
18	202.77.12	18-30	Male	High scho	Employee	Below 1yr	Lorry (11?	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
19	10.42.0.42	31-50	Male	Junior hig	Employee	1-2yr	Automobil	Owner	Unknown	No defect	Tangent ro Asphalt ro	Wet or dan	Daylight	
20	151.101.2	18-30	Male	Junior hig	Employee	Above 10yr	Other	Owner	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
21	10.42.0.42	31-50	Male	Junior hig	Owner	Above 10yr	Automobil	Owner	2-5yrs	No defect	Tangent ro Other	Dry	Daylight	
22	172.217.1	31-50	Male	Junior hig	Employee	Above 10yr	Automobil	Government	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
23	172.217.1	18-30	Male	Junior hig	Employee	No Licence	Automobil	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
24	172.217.1	18-30	Male	Junior hig	Employee	1-2yr	Automobil	Owner	5-10yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
25	10.42.0.21	Over 51	Male	Illiterate	Owner	1-2yr	Lorry (41?	Owner	2-5yrs	No defect	Steep grad Asphalt ro	Wet or dan	Darkness	
26	172.217.1	31-50	Male	Junior hig	Owner	Below 1yr	Other	Owner	1-2yr	No defect	Tangent ro Asphalt ro	Dry	Daylight	
27	10.42.0.21	31-50	Male	Junior hig	Owner	1-2yr	Pick up up	Organizati	2-5yrs	No defect	Tangent ro Asphalt ro	Dry	Daylight	
28	211.144.3	Over 51	Male	Junior hig	Owner	2-5yr	Pick up up	Government	1-2yr	No defect	Tangent ro Asphalt ro	Wet or dan	Darkness	
29	10.42.0.15	Unknown	Female	Unknown	Owner	5-10yr	Automobil	Owner	Unknown	No defect	Tangent ro Asphalt ro	Wet or dan	Darkness	
30	222.73.28	31-50	Male	Unknown	Owner	Above 10yr	Long lorry	Owner	2-5yrs	No defect	Tangent road with flat	Dry	Darkness	
31	172.217.1	31-50	Male	Junior hig	Owner	2-5yr	Automobil	Owner	Unknown	No defect	Tangent ro Other	Dry	Darkness	
32	10.42.0.21	18-30	Male	Elementar	Owner	Above 10yr	Public (13?	Government	1-2yr	No defect	Tangent ro Asphalt ro	Dry	Daylight	
33	198.105.2	18-30	Male	Writing &	Owner	5-10yr	Lorry (41?	100Q	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
34	157.240.1	18-30	Male	Owner	Owner	5-10yr	Public (13?	Organizati	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	
35	10.42.0.42	18-30	Male	Junior hig	Owner	5-10yr	Other	Owner	Above 10yr	No defect	Steep grad Asphalt ro	Dry	Daylight	
36	182.22.31	18-30	Male	Junior hig	Owner	Below 1yr	Lorry (41?	Owner	Unknown	No defect	Tangent ro Asphalt ro	Dry	Daylight	

Home Page



Admin Login



Admin Dashboard

Predicting Bike Usage and Optimizing Operations at Repair Shops in Bike Sharing Systems

Train & Test Bike Data Sets View Trained and Tested Bike Datasets Accuracy in Bar Chart View Trained and Tested Bike Datasets Accuracy Results View Bike Usage and Repair Time Type Find Bike Usage and Repair Time Type Ratio

Download Predicted Datasets View Bike Usage and Repair Time Type Ratio Results View All Remote Users Logout

VIEW ALL REMOTE USERS !!!

USER NAME	EMAIL	Gender	Address	Mob No	Country	State	City
Rajesh	Rajesh123@gmail.com	Male	#8928,4th Cross,Rajajinagar	9535866270	India	Karnataka	Bangalore
Manjunath	tmksmanju14@gmail.com	Male	#8928,4th Cross,Vijayanagar	9535866270	India	Karnataka	Bangalore
sai	sai@gmail.com	Male	kdp	9849294832	india	ap	kdp
eswar	eswarmatidevaraju95@gmail.com	Male	utukur	8790594403	india	ap	rjpt
eswar	eswarmatidevaraju@gmail.com	Male		8790594403	india	ap	kdp
kiran	kiran@gmail.com	Male		8790594403	india	ap	kdp
kiran	kiran@gmail.com	Male		8790594403	india	ap	kdp
raju	raju@gmail.com	Male		7689054	india	ap	kdp

Training and Accuracy

```
C:\Windows\System32\cmd.e  X + v
accuracy          0.48      4805
macro avg         0.16      0.33      0.22      4805
weighted avg      0.23      0.48      0.31      4805

SVM
47.92924037460978
CLASSIFICATION REPORT
      precision    recall  f1-score   support

     0       0.00      0.00      0.00     1054
     1       0.48      1.00      0.65     2303
     2       0.00      0.00      0.00     1448

 accuracy          0.48      4805
macro avg         0.16      0.33      0.22      4805
weighted avg      0.23      0.48      0.31      4805

CONFUSION MATRIX
[[ 0 1054  0]
 [ 0 2303  0]
 [ 0 1448  0]]

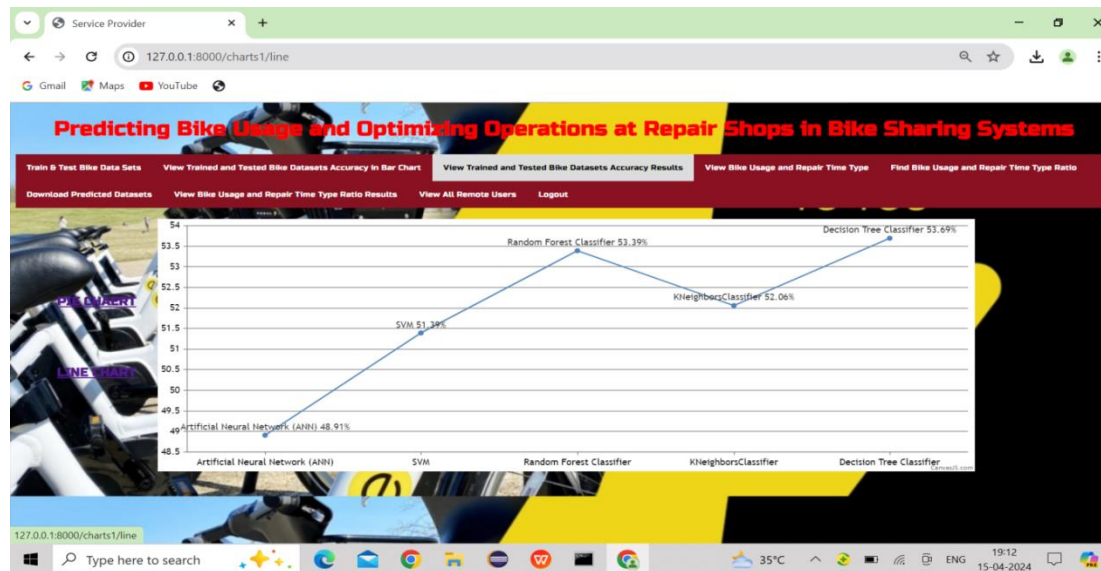
XGBoost
ACCURACY
47.92924037460978
CLASSIFICATION REPORT
      precision    recall  f1-score   support

     0       0.00      0.00      0.00     1054
     1       0.48      1.00      0.65     2303
     2       0.00      0.00      0.00     1448

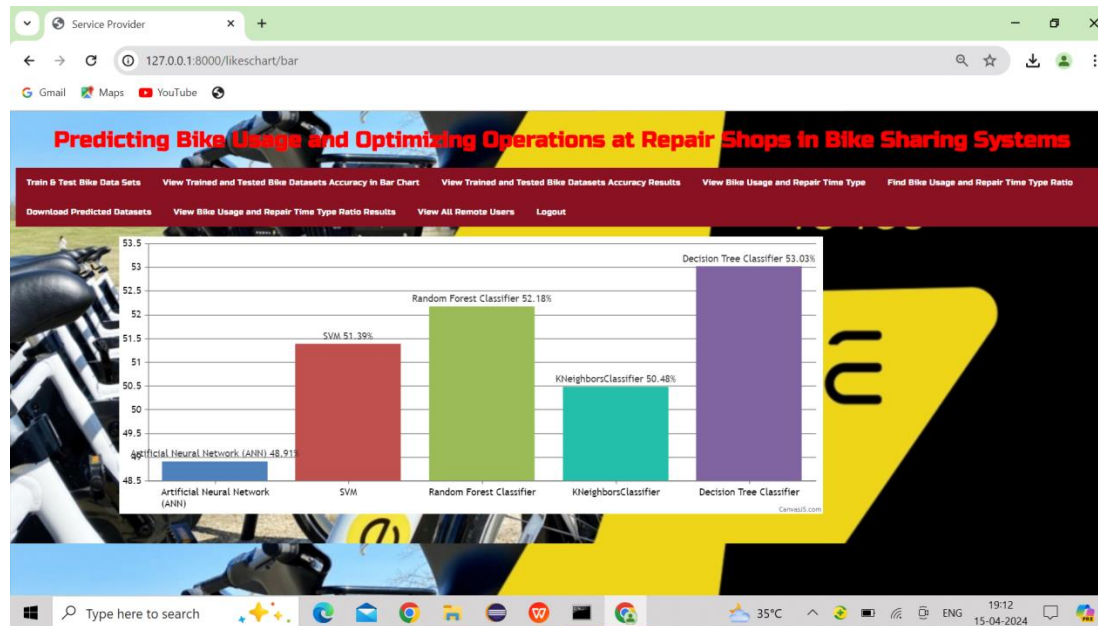
 accuracy          0.48      4805
macro avg         0.16      0.33      0.22      4805
weighted avg      0.23      0.48      0.31      4805

CONFUSION MATRIX
[[ 0 1054  0]
 [ 0 2303  0]
 [ 0 1448  0]]
Random Forest Classifier
ACCURACY
```

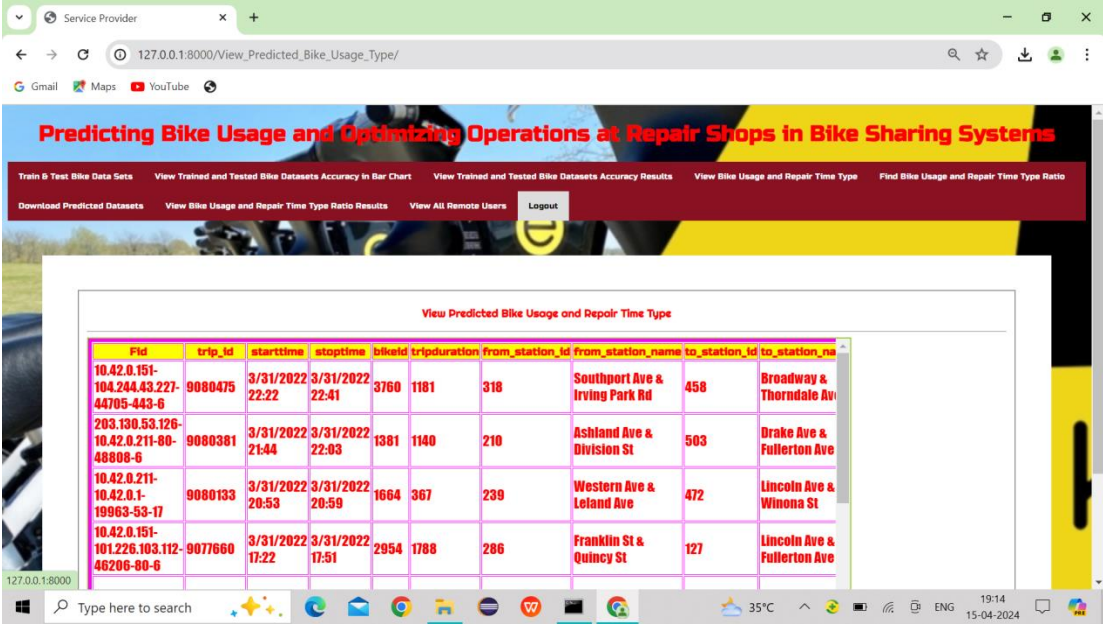
ML Accuracy in Line Chart



Accuracy in Barchat



Predicted Results



Predicting Bike Usage and Optimizing Operations at Repair Shops in Bike Sharing Systems

Train & Test Bike Data Sets View Trained and Tested Bike Datasets Accuracy in Bar Chart View Trained and Tested Bike Datasets Accuracy Results View Bike Usage and Repair Time Type Find Bike Usage and Repair Time Type Ratio

Download Predicted Datasets View Bike Usage and Repair Time Type Ratio Results View All Remote Users Logout

View Predicted Bike Usage and Repair Time Type

Fld	trip_id	starttime	stoptime	bikeid	tripduration	from_station_id	from_station_name	to_station_id	to_station_name
10.42.0.151-104.244.43.227-44705-443-6	9080475	3/31/2022 22:22	3/31/2022 22:41	3760	1181	318	Southport Ave & Irving Park Rd	458	Broadway & Thorndale Ave
203.130.53.126-10.42.0.211-90-48808-6	9080381	3/31/2022 21:44	3/31/2022 22:03	1381	1140	210	Ashland Ave & Division St	503	Drake Ave & Fullerton Ave
10.42.0.211-10.42.0.1-19963-53-17	9080133	3/31/2022 20:53	3/31/2022 20:59	1664	367	239	Western Ave & Leland Ave	472	Lincoln Ave & Winona St
10.42.0.151-101.226.103.112-46206-80-6	9077660	3/31/2022 17:22	3/31/2022 17:51	2954	1788	286	Franklin St & Quincy St	127	Lincoln Ave & Fullerton Ave

Data Input for Prediction

Remote User

127.0.0.1:8000/predict_bike_usage_type/

Gmail Maps YouTube

PREDICT BIKE USAGE AND REPAIR TIME TYPE !!!

ENTER BIKE USAGE DATASET DETAILS HERE !!!

Enter fid	100 11 132 221-10 42 0 15	Enter trip_id	9080545
Enter starttime	3/31/2022 23:30	Enter stoptime	4/31/2022 23:46
Enter bikeid	1234	Enter tripduration	926
Enter from_station_id	156	Enter from_station_name	Clark St & Wellington Ave
Enter to_station_id	166	Enter to_station_name	Ashland Ave & Wrightwood
Enter usertype	Subscriber	Enter gender	Male
Enter birthyear	2002		

Predict

Predicted Bike Usage and Repair Time Type :

Type here to search

35°C

19:18
15-04-2024

Prediction Results

The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8000/predict_bike_usage_type/`. The page title is "PREDICT BIKE USAGE AND REPAIR TIME TYPE III". The main content area is titled "ENTER BIKE USAGE DATASET DETAILS HERE III" and contains two columns of input fields. The left column includes fields for "Enter Fid", "Enter starttime", "Enter bikeid", "Enter from_station_id", "Enter to_station_id", "Enter usertype", and "Enter birthyear". The right column includes fields for "Enter trip_id", "Enter stoptime", "Enter tripduration", "Enter from_station_name", "Enter to_station_name", and "Enter gender" (with a dropdown menu). A "Predict" button is located below the input fields. Below the prediction area, there is a section titled "Predicted Bike Usage and Repair Time Type =" followed by a "More" link. The Windows taskbar at the bottom shows the search bar, various application icons, and system information including the temperature (35°C), date (15-04-2024), and time (19:26).

PREDICT BIKE USAGE AND REPAIR TIME TYPE III

ENTER BIKE USAGE DATASET DETAILS HERE III

Enter Fid		Enter trip_id	
Enter starttime		Enter stoptime	
Enter bikeid		Enter tripduration	
Enter from_station_id		Enter from_station_name	
Enter to_station_id		Enter to_station_name	
Enter usertype		Enter gender	--Select--
Enter birthyear			

Predict

Predicted Bike Usage and Repair Time Type =

More

Conclusion

- The proposed approach demonstrates that combining machine learning with supply chain network design significantly improves bike-sharing system performance.
- The model enables faster repair decisions, reduces lead time, and increases throughput. Operational flexibility proves to be a key factor in handling big data challenges effectively.

Paper publication details:

4/6/26, 1:15 PM

KSRM College of Engineering Mail - Fwd: 2026 IEEE - Conference on Latest Innovations in Computing and Knowledge 2026 (C...



VENNAPUSA GOWTHAMI <229y1a3963@ksrmce.ac.in>

Fwd: 2026 IEEE - Conference on Latest Innovations in Computing and Knowledge 2026 (CLICK 2026) : Submission CLICK1656 has been created.

1 message

Gowthami Vennapusa <gowthamivennapusa9@gmail.com>
To: 229y1a3963@ksrmce.ac.in

Mon, Apr 6, 2026 at 1:11 PM

----- Forwarded message -----

From: <ieee@hindustanuniv.ac.in>
Date: Sun, Mar 29, 2026, 2:46 PM
Subject: 2026 IEEE - Conference on Latest Innovations in Computing and Knowledge 2026 (CLICK 2026) : Submission CLICK1656 has been created.
To: <gowthamivennapusa9@gmail.com>

Dear V.Gowthami,

Thank you for submitting your article to **2026 IEEE - Conference on Latest Innovations in Computing and Knowledge 2026 (CLICK 2026)**,

The following submission has been created:

Track Name: Data - Driven Business AI
Paper ID: CLICK1656
Paper Title: Using AI - Predicting Bike Usage and Optimizing Operations at Repair in Bike Sharing System
Created on: Sun Mar 29 2026 14:46:31 GMT+0530 (India Standard Time)

Your submission is currently undergoing the review process, and we will notify you of the its status once we receive the review report.

Regards,
Conference Chairs.

With Regards,
Conference - Chair
CLICK - 2026.



"TO MAKE EVERY MAN A SUCCESS AND NO MAN A FAILURE"
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References

- [1] E. Alpaydin, Introduction to Machine Learning. Cambridge, MA, USA: MIT Press, 2020.
- [2] R. Alvarez-Valdes, J. M. Belenguer, E. Benavent, J. D. Bermudez, F. Muñoz, E. Vercher, and F. Verdejo, “Optimizing the level of service quality of a bike-sharing system,” *Omega*, vol. 62, pp. 163–175, Jul. 2016.
- [3] C. Alzaman, Z.-H. Zhang, and A. Diabat, “Supply chain network design with direct and indirect production costs: Hybrid gradient and local search based heuristics,” *Int. J. Prod. Econ.*, vol. 203, pp. 203–215, Sep. 2018.
- [4] O. C. Arqué. (Jul. 7, 2020). Demand Forecast Model For A Bicycle Sharing Service. [Online]. Available: <https://upcommons>.

THANK YOU